

Computer Vision & Cognitive Systems

SCQ5109806 - LM CS,DS,CYB,PD

Local features: visual (logo) recognition contest

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DIPARTIMENTO MATEMATICA
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SIFT matching

- An example:



Visual Recognition Contest

- **Task:** instance-based object recognition
  - Test different combinations of local visual features and image matching strategies
  - Participate by submitting your solutions and check the leaderboard/results (links on Moodle)



- **Query images**: 4 logos
- **Dataset** (testset): 110 images
- **Evaluation metrics:** precision, recall, mean avg precision (mAP)

Visual Recognition Contest

- Deadline: November 8, 2026 - 11:00pm
- You can participate individually or as a group (2 persons)
- In your entries (max 3 submissions) as username put full name(s) and add a short description of what you have done in the note field

- Top-ranked groups shall present their approach (~2 min)



- **Query images**: 4 logos
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Evaluation metrics

- **Precision** and **Recall** are the standard metrics used for evaluating information retrieval systems
  - Precision = TP / (TP + FP)
  - Recall = TP / (TP + FN)

- An example:

		+ In Result	- Not in result
+	Relevant	30 (TP)	20 (FN)
-	Not Relevant	10 (FP)	40 (TN)

Groundtruth

Output

Prec = 30 / (30 +10) = 0.75

Rec = 30 / (30 +20) = 0.60

Evaluation metrics

- It is common to plot *Precision-Recall curves* (or ROC) for different cut-off points of a parameter

- For systems that return a ranked list *Average Precision (AP)* is the most common evaluation metric
  - We can plot precision as a function of recall r: i.e. $p(r)$
  - AP is the average value of $p(r)$ over the interval from r=0 to r=1 (i.e. it computes the area under the curve)

$$\text{AveP} = \frac{\sum_{k=1}^{n} (P(k) \times \text{rel}(k))}{\text{number of relevant documents}}$$

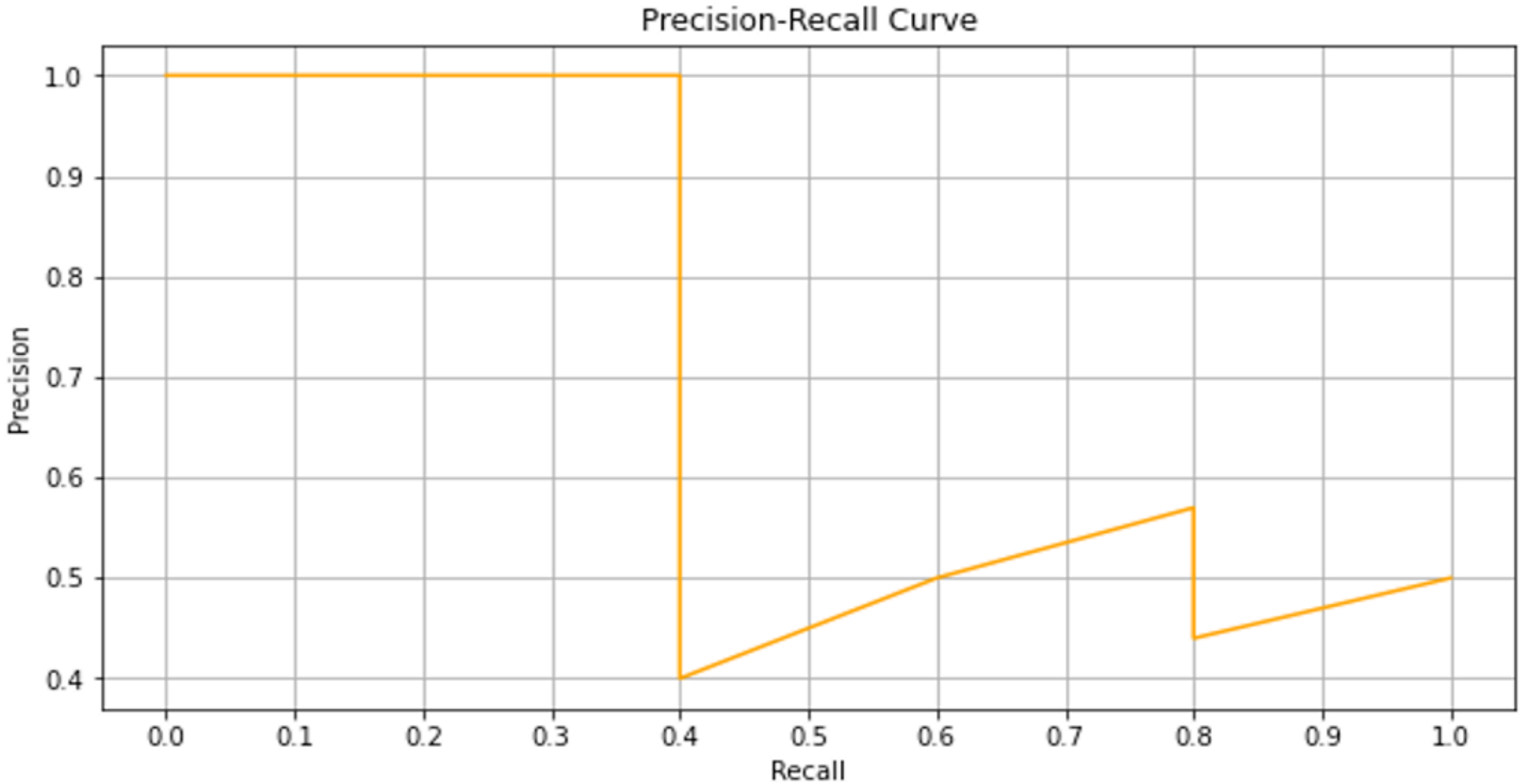
Evaluation metrics

- AP, an example:
  - Let's consider a dataset of 10 images (5 containing the searched logo, e.g. *Coca Cola*):

Rank	Score	Correct?	Precision	Recall
1	210	TRUE	1.0	0.2
2	150	TRUE	1.0	0.4
3	147	FALSE	0.67	0.4
4	135	FALSE	0.5	0.4
5	122	FALSE	0.4	0.4
6	80	TRUE	0.5	0.6
7	75	TRUE	0.57	0.8
8	45	FALSE	0.5	0.8
9	23	FALSE	0.44	0.8
10	12	TRUE	0.5	1.0

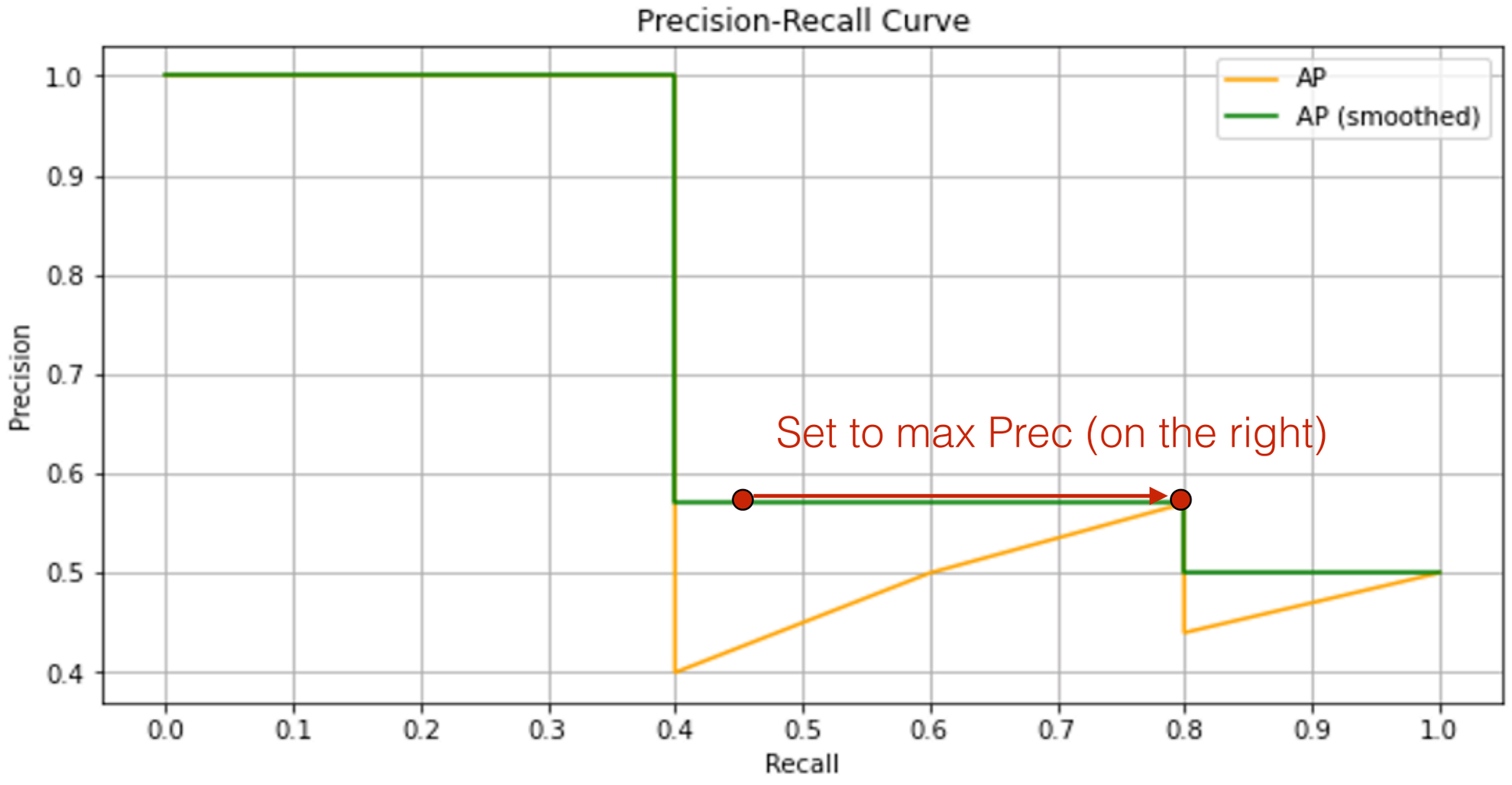
Evaluation metrics

- From AP to Interpolated AP (through an example):



Evaluation metrics

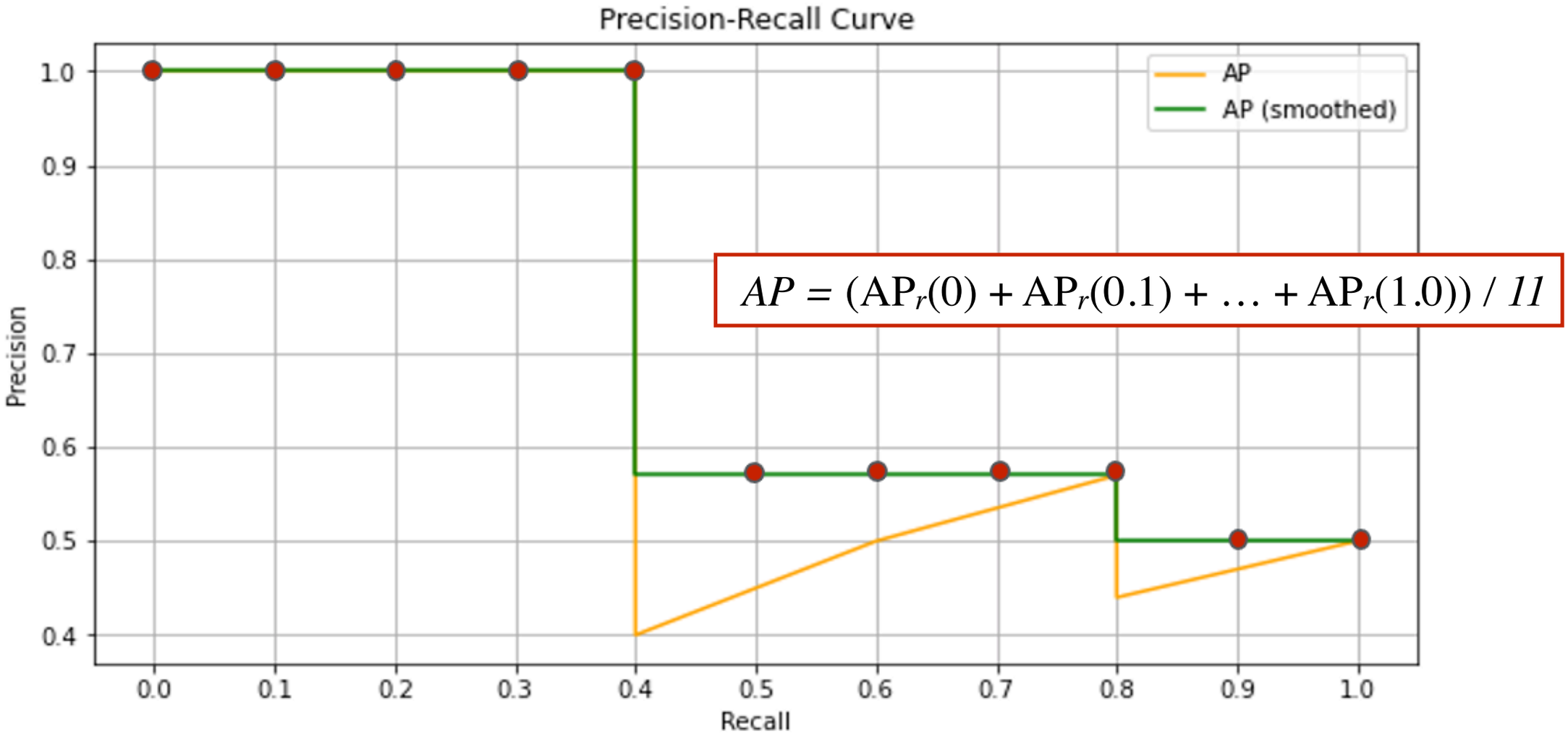
- From AP to Interpolated AP (through an example):



Evaluation metrics

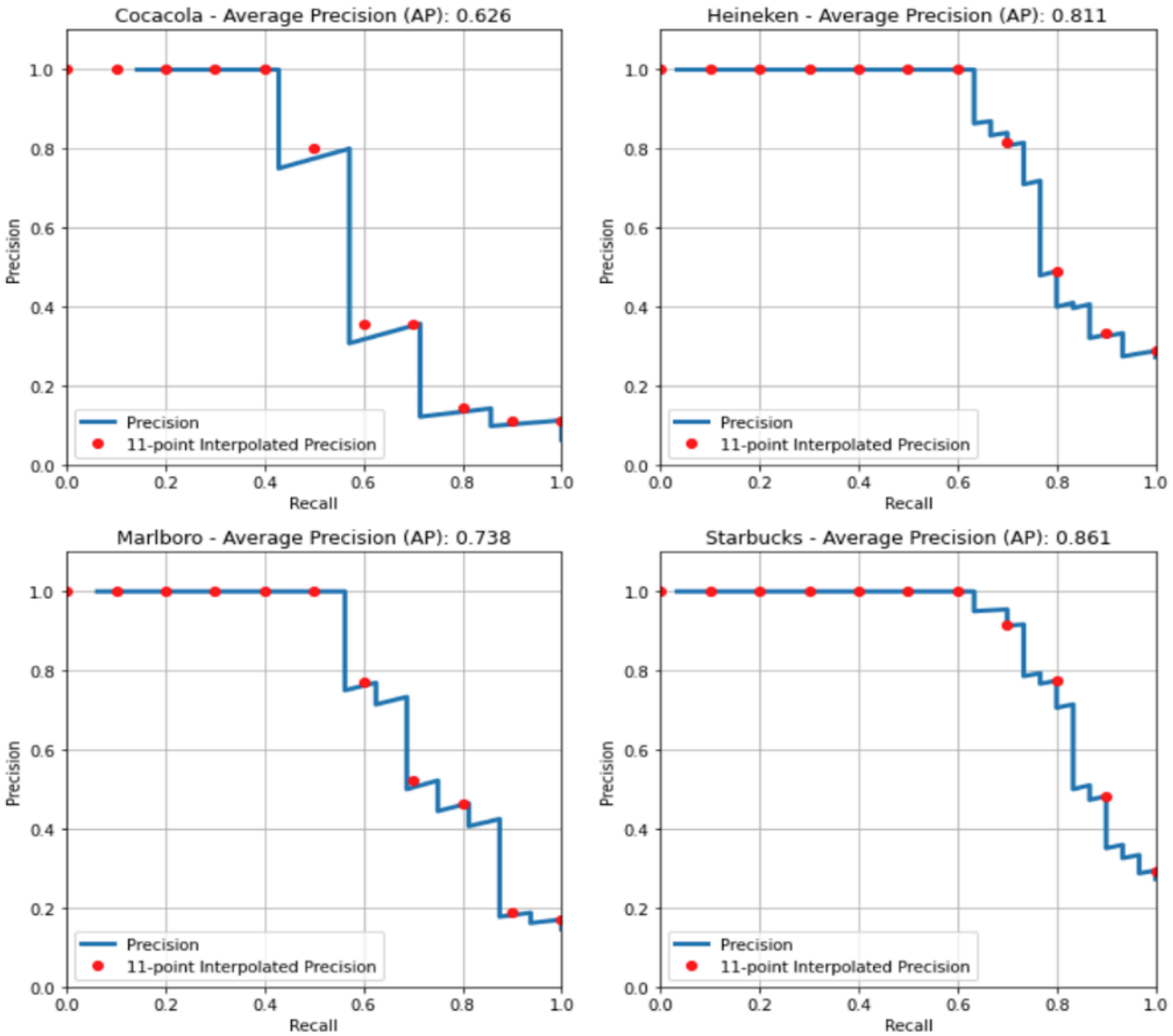
- From AP to Interpolated AP (through an example):

  Discretize the recall values from 0 to 1.0 into 11 points: 0, 0.1, 0.2, ….
  Next, compute the average of ma precision value for these recall values



Visual Recognition Contest

- Our baseline:



Our pipeline:

- Standard SIFT (DoG, 128-d descriptors)
- SIFT Matching
- Threshold on #matches

(Interpolated AP)

```
Average Precision (AP):

Cocacola   ===>  0.626
Heineken   ===>  0.811
Marlboro   ===>  0.738
Starbucks  ===>  0.861

Mean Average Precision (mAP) ===> 0.759
```

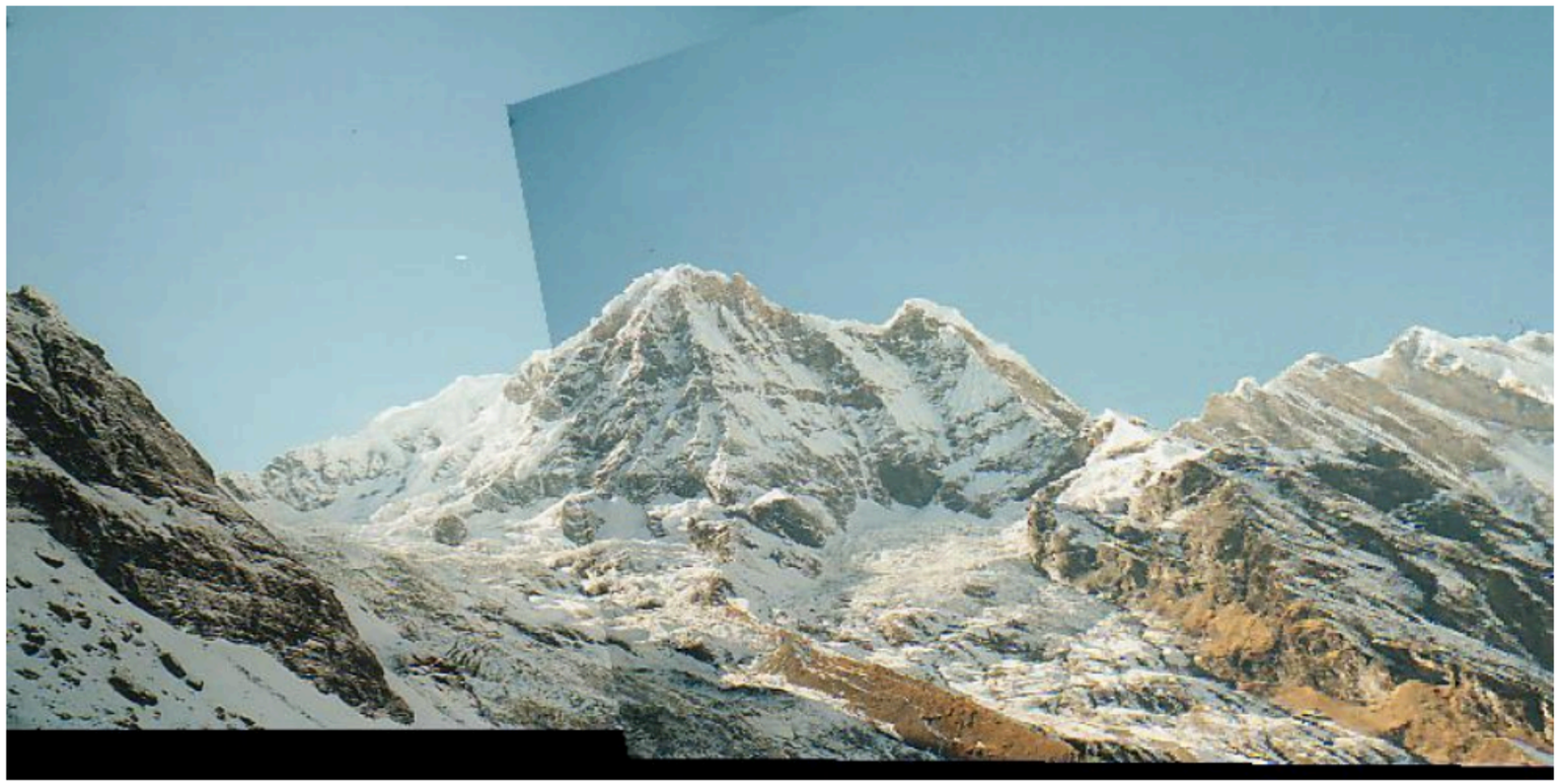
Visual Recognition Contest

- A few advice:
  - Check and try different parameters of the SIFT algorithm (both detection, e.g. octaves, and description)
  - Try different local visual features (e.g. SURF, ORB, …) or different combinations of detectors + descriptors

Feature Detector	Corner	Blob	Region	Rotation invariant	Scale invariant	Affine invariant	Repeatability	Localization accuracy	Robustness	Efficiency
Harris	√			√			+++	+++	+++	++
Hessian		√		√			++	++	++	+
SUSAN	√			√			++	++	++	+++
Harris-Laplace	√	(√)		√	√		+++	+++	++	+
Hessian-Laplace	(√)	√		√	√		+++	+++	+++	+
DoG	(√)	√		√	√		++	++	++	++
SURF	(√)	√		√	√		++	++	++	+++
Harris-Affine	√	(√)		√	√	√	+++	+++	++	++
Hessian-Affine	(√)	√		√	√	√	+++	+++	+++	++
Salient Regions	(√)	√		√	√	(√)	+	+	++	+
Edge-based	√			√	√	√	+++	+++	+	+
MSER			√	√	√	√	+++	+++	++	+++
Intensity-based			√	√	√	√	++	++	++	++
Superpixels			√	√	(√)	(√)	+	+	+	+

Visual Recognition Contest

- A few advice:
  - Try different features and “classification” strategies…



Step 1: detect keypoints

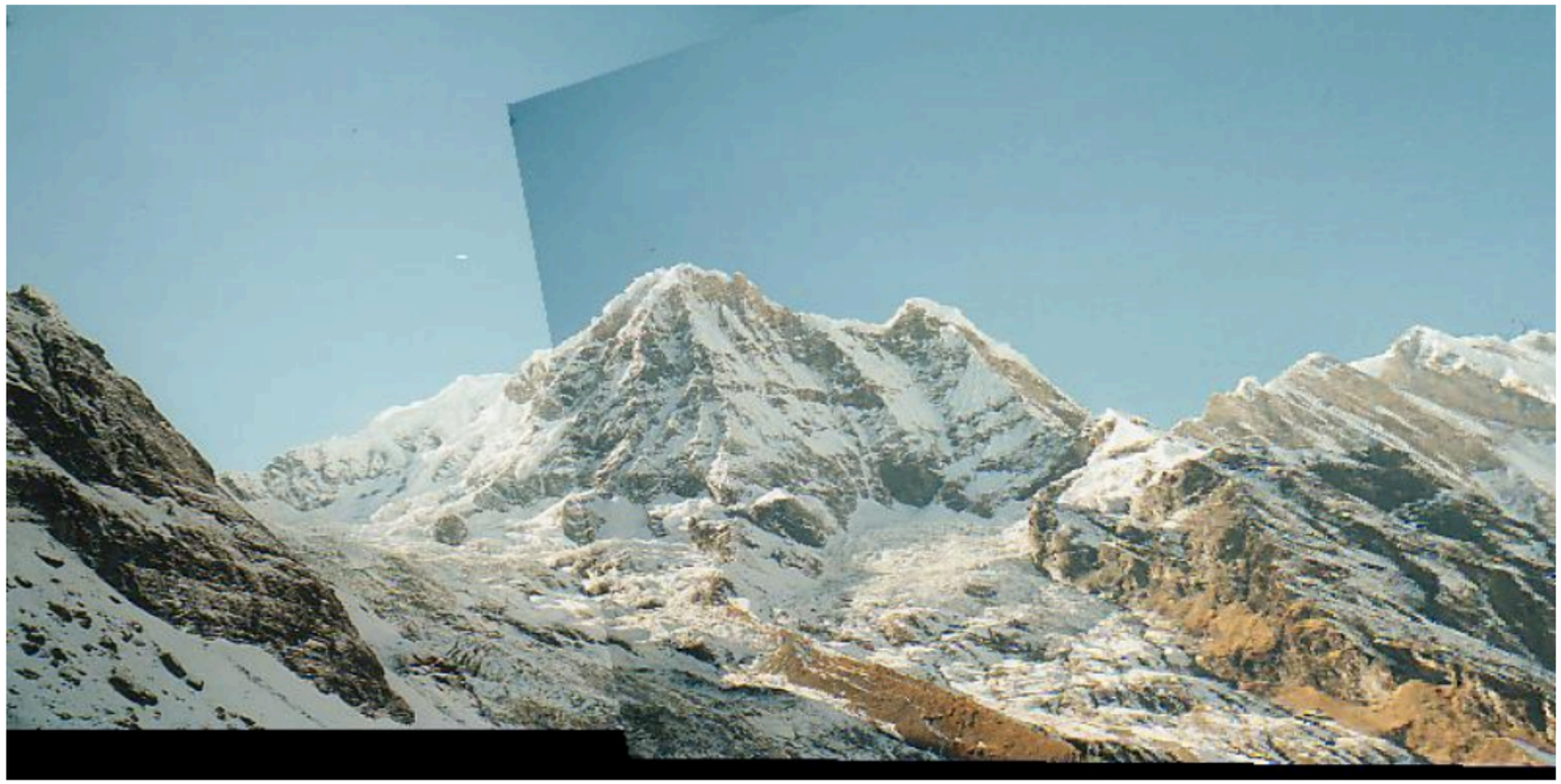
Step 2: build keypoints descriptors

Step 3: match keypoint features

Step 4: align images (i.e. fitting the transformation)

Visual Recognition Contest

- A few advice:
  - Try different features and “classification” strategies… as well as better image matching approaches



Step 1: detect keypoints

Step 2: build keypoints descriptors

Step 3: match keypoint features

Step 4: align images (i.e. fitting the transformation)

Contact

- **Office:** Torre Archimede, room 6CD3
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